Research Dossier — Part 2: Technical Companion

Algorithms · Hypothesis Failure · Theory vs Practice · Novelty · Success Rate · PhysioNet

Hierarchical Federated Learning for Privacy-Aware, Low-Latency Multimodal IoT Systems (PP-HFM)
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1 — ALGORITHMS USED, AND WHY

Algorithm	Where used	Why this one
FedAvg	Aggregation at edge and cloud	The standard. Simple, well understood, and the correct baseline. Anything more exotic would need to justify itself against it.
DP-SGD (Opacus)	Private training of the fusion head	The only mechanism giving a formal, composable guarantee. Everything else is heuristic.
Patient-level DP-SGD (our variant)	The core privacy mechanism	Standard DP-SGD clips *per sample*. Our sample is a 10-minute window and every patient gives about 8 of them. Clipping per window protects windows, not people. We group by patient first, then clip.
Top-*k* sparsification (20 %)	Gradient compression	Sends only the 20 % largest updates. 5× reduction. Cheap and effective.
8-bit quantisation	Gradient compression	Another 4× on top of sparsification. Combined 20×.
Error feedback	Compression accumulator	Compression throws information away. Error feedback stores what was dropped and adds it back next round, so accuracy is not lost.
Head-only DP fine-tuning	The training recipe	Freeze the encoders, privatise only the fusion head (26 % of parameters). This is what made $\varepsilon \leq 1$ work at all.
Multi-scale 1D-CNN + BiGRU + attention	Signal encoder (modality A)	Captures patterns at several time scales. Uses DP-compatible layers — no BatchNorm, which breaks DP-SGD.
MobileNetV3 (ImageNet-pretrained)	Image encoder	Small enough for edge deployment, and pretrained on public data so freezing it is privacy-valid.
I3D two-stream features	Video modality (WEAR)	Standard egocentric video representation. Used as given, not trained.
Conformal prediction (LAC score)	Uncertainty at inference	Gives a coverage guarantee: the true label is in the predicted set with probability $\geq 1-\alpha$. Important for clinical use.
DP quantile release	Conformal calibration	Releasing the calibration threshold leaks information. We pay for it: $\varepsilon = 0.20$ charged to the same budget. Most papers omit this.
LiRA (likelihood-ratio attack)	Privacy audit	The strongest membership inference attack available. A weak attack producing a null proves nothing.
Integrated Gradients	Feature attribution	Shows what the model actually uses. Revealed that missingness masks carry 22 % of attribution mass.
RDP accounting (extended orders)	Privacy budget tracking	Opacus's default orders stop at $\alpha = 63$, which puts a floor of ≈0.19 on ε. We extended the orders so tight budgets are reachable.
Tried and dropped: ASWA aggregation (contributed +0.0005 AUROC — nothing) · SADP adaptive privacy (worse than uniform DP at equal ε) · FedKD-E distillation (not freeze-aware, would leak) · MOON, FedProx (baselines only) · INT8 quantisation (slower than FP32)		

2 — WHY THE HYPOTHESIS FAILED, AND HOW TO GET IT BACK

2.1 What we predicted

If you protect only *records*, an attacker should be able to tell whether a *patient* was in the training set. We predicted an attack success of about **0.70** against record-level DP and about **0.50** (chance) against patient-level DP. A gap of roughly **0.20**.

2.2 What we measured

Record-level DP: **0.5477**. Patient-level DP: **0.5224**. Gap: **+0.0325**, and the confidence interval includes zero. All four attack variants gave the same answer.

2.3 Why it failed — the real reason

Not because the attack was weak. We proved that first: the positive control, on a model known to memorise, reached **0.6277** — it cleared the gate we set before running.

The honest reason is this. **DP-SGD at $\varepsilon = 1$ is already very strong.** The noise ($\sigma = 11.33$) is large enough that even record-level DP leaves almost no membership signal to find. The theoretical gap between record-level and patient-level protection is real, but at this budget both arms are pushed so close to chance that the difference between them is smaller than we can measure with 5 seeds.

A supporting clue confirms this. On the patient arm, LiRA actually performed *worse* than a plain loss threshold (−0.0071). That is what a defence with nothing left to leak looks like — not a failing attack.

> **Say it this way, always**
>
> The **large** predicted effect is refuted. A **small** effect is not excluded — the interval is ±0.060 wide at n = 5. Never write “there is no effect”.

2.4 What we would need to get it back

#	What to change	Why it would help	Cost
1	**Attack at a weaker privacy budget** — ε = 4, 8, 16 instead of 1	At ε = 1 both arms are near chance, so there is no room for a gap. Weaker DP leaves signal, and the record-vs-patient difference has space to show. This is the most likely route.	Low
2	**Increase seeds** from 5 to 20–30	The interval half-width is ±0.060. The measured gap is +0.0325. More seeds would resolve whether that small gap is real.	Medium
3	**Use a dataset with higher *k***	VitalDB gives about 8 windows per patient. The more records a patient contributes, the bigger the theoretical gap. A dataset with 50–100 records per patient should show it more clearly.	Needs eICU
4	**Attack the worst-case patient, not the average**	Average AUC hides the tail. Report TPR at 0.1 % FPR — the confident identifications that are the real clinical disclosure. The machinery already computes this.	Low
5	**Attack the head-only recipe directly**	We already know head-only training memorises harder (0.6277 vs 0.5122). That is where leakage lives. This is the finding that survived.	Done

Route 1 is the recommendation. The hypothesis may not be wrong — it may simply have been tested at a budget where nothing is visible.

3 — THEORY VS PRACTICE

3.1 What we achieved in theory

Theoretical result	Status	Meaning in one line
Patient-level guarantee holds by construction	**PROVEN**	Grouping gradients before clipping bounds one patient's influence, whatever number of records they contribute.
Record-level ε ≠ patient-level ε	**PROVEN**	The two are different guarantees. This is analytic and survives the failed attack.
Price of upgrading the unit = 1.77× σ	**DERIVED**	Making the guarantee patient-level costs 1.77× more noise at the same budget.
Group privacy conversion is unusable here	**SHOWN**	At 8 records per patient, converting a record-level ε = 1 gives a number that guarantees nothing.
Composed budget = training + calibration	**CORRECT**	ε = 0.7705 + 0.2000 = 0.9705. Most papers report only the first term.
Conformal coverage guarantee	**HOLDS**	P(true label in predicted set) ≥ 1 − α, and its release cost is paid for.
Cost is a function of *k*, zero at *k* = 1	**PREDICTED & CONFIRMED**	Theory says the two units coincide at *k* = 1. Measurement agrees (p = 0.75). Theory and experiment match.
Record-level DP is exploitable by patient MIA	**PREDICTED, NOT OBSERVED**	The theory is sound; the effect is not measurable at ε = 1.

3.2 What we achieved in practice

Practical result	Measured	Verdict
Uplink reduction (VitalDB / WEAR)	90.82 % / 96.08 %	**WORKS**
Energy saving	72.30 % / 76.84 %	**WORKS**
Inference latency P95	4.58 ms / 1.22 ms	**WORKS**
Update reliability	100 % / 99.94 %	**WORKS**
Privacy budget achieved	ε = 0.9705 / 0.9656	**WORKS**
Accuracy cost at ε = 1 (VitalDB)	−0.92 pp	**WORKS**
Multimodal fusion gain (VitalDB)	+0.0370 AUROC	**WORKS**
k-scaling law measured	24/24 configs	**WORKS**
Accuracy cost at ε = 1 (WEAR)	1.967 pp, CI [1.76, 2.17]	**UNRESOLVED**
Patient MIA vulnerability	Δ CI includes zero	**FAILED**
Adaptive DP (SADP)	σ 11.3 → 70	**FAILED**
INT8 edge quantisation	1.9× slower on CPU	**FAILED**
Fusion gain on WEAR	+0.0004 (ceiling)	**NO HEADROOM**
ASWA aggregation	+0.0005 AUROC	**USELESS**
End-to-end valid ε (public pretraining)	priced at −0.022 AUROC	**NOT CLOSED**

The short version

Theory: almost everything held. The one thing that did not was a prediction about what an attacker could do, not about the mechanism itself.

Practice: every system objective passed with a wide margin. The failures were all in the optional extras — adaptive privacy, INT8, a fancy aggregator — none of which the framework depends on.

4 — BEST AND UNIQUE FINDINGS

Rank	Finding	Why it is unique
1	**The cost of patient-level privacy is a function of *k*, not a fixed number.** Gap 0 → +0.0507 → +0.0832 → +0.0866 AUPRC as *k* goes 1 → 2 → 4 → 8.	Every other paper reports privacy cost as one number. We show it is a curve, and which variable controls it. It also carries its own control: at *k* = 1 the two guarantees are identical and the arms must collapse — they do.
2	**The gap widens because the record arm improves, not because the patient arm degrades.** Patient arm is flat (0.4967–0.5176 at every *k*); record arm rises (0.5015 → 0.5955).	This is the mechanism, and it is counter-intuitive. Extra records per patient buy the patient arm nothing, because the noise scales with the patient's total contribution.
3	**Head-only DP fine-tuning memorises members far harder than full-model training.** 0.6277 vs 0.5122 under LiRA.	This is a warning about the very technique the field is adopting for efficiency — and about our own recipe. Nobody has reported it.
4	**At ε = 1 the noise was never the problem — the optimisation geometry was.** σ = 11.3281 identical before and after; AUC moved 0.562 → 0.8742.	Reframes the entire "DP destroys utility" complaint. A near-linear probe over frozen features tolerates noise that end-to-end training cannot.
5	**Differential privacy shifts the model's attention toward missingness.** Mask attribution share: full_dp 28.2 % > patient 22.0 % > full_nodp 19.9 % > head_nodp 15.8 %.	An unexpected side effect of privacy: under noise the model leans more on *which measurements exist* than on their values. Nobody has measured this.
6	**A negative result that is actually trustworthy.** Patient MIA null, with a positive control that passed its pre-registered gate at 0.6277.	Most negative results in this area are unpublishable because the attack might just be weak. Ours cannot be dismissed that way.
8	**The difficulty floor: MIA numbers in this setting must be read against a confound, not against chance.** Ranking patients by shadow difficulty alone — with no target model — scores 0.5252 to 0.5743, while the attacks themselves reach only 0.5465–0.5477.	Membership attacks are conventionally reported against 0.5. Where members come from the train split and non-members from the test split, that is the wrong reference: the split's own difficulty imbalance is credited to the attacker. We measure the floor and report every attack as a margin over it. At ε = 16 the attack falls *below* its floor — no signal at all.
7	**Some IoT tasks do not need fusion at all.** WEAR's best single modality reaches 0.9976 AUROC; fusion adds +0.0004.	An honest limit on the multimodal argument that the multimodal FL literature never states.

5 — SUCCESS RATE OF THE RESEARCH

Category	Achieved	Rate	Note
Engineering objectives (both datasets)	13 of 14	**93 %**	Only WEAR O2b unresolved, and deliberately so
Original four objectives (O1–O4)	4 of 4	**100 %**	All exceeded, most by a wide margin
Project phases	5 of 6	**83 %**	Phase 6 code-complete, runs partly pending
Phase 5 deliverables	17 of 17	**100 %**	
Scientific claims measured	8 of 12	**67 %**	1 refuted, 2 never run, 1 evidence-only
Major hypotheses	1 of 2	**50 %**	*k*-law succeeded; patient MIA refuted
Optional mechanisms we designed	0 of 3	**0 %**	SADP, ASWA, FedKD-E all dropped
External validation	0 of 1	**0 %**	Blocked on eICU access

Overall, honestly

Engineering: ~95 % successful. The system does everything it was supposed to do, with margins of 2× to 20× on the targets.

Science: ~60 % successful. One hypothesis died, one succeeded on a single dataset, and one original mechanism has still not been tested.

Combined, a fair figure is 75–80 %. The remaining 20–25 % is almost entirely the external validation that eICU would supply.

Rating against publication: currently around **8.8/10** in research strength, with roughly a **60–75 %** chance of a strong publication outcome. A successful eICU validation would move that to roughly **9.2–9.4/10** and **75–90 %**.

6 — WHAT DOES NOT EXIST ANYWHERE ELSE

This is the list of things we could not find in any paper we studied. It is the answer to “what is new?”

#	Does not exist in the literature	Closest existing work, and what it stops short of
1	**The privacy cost measured as a curve in records-per-individual.**	Nobody has done this. User-level DP papers (McMahan 2018) report one number for one dataset. The controlling variable has never been isolated.
2	**A hierarchical FL system that carries a formal patient-level guarantee.**	HED-FL, HCEF and Mhaisen et al. are hierarchical with **no privacy guarantee at all**. FL2DP and RoPPFL have privacy but are **not hierarchical**. The two lines have never met.
3	**Conformal calibration cost composed into the privacy budget.**	Conformal FL papers release the calibration quantile for free. Smith (2011) gives the DP quantile mechanism, but nobody charges it to the same budget as training. We do: 0.7705 + 0.2000.
4	**Patient-level membership inference audit with a passing positive control.**	MIA audits in FL are conducted at record level — the quantity a record-level guarantee already protects, so the least informative test. Almost none report a positive control, so their nulls cannot be trusted.
5	**The finding that parameter-efficient DP fine-tuning increases memorisation.**	Li et al. (2021) and Yu et al. (2021) propose head-only DP fine-tuning for efficiency. Neither measures whether it leaks more. It does.
6	**Noise calibrated against every client simultaneously in a hierarchy.**	Standard practice calibrates against a representative client. That silently understates the budget when an edge tier changes how often each client is sampled. The failure is invisible in a training log.
7	**DP shifting attention toward missingness.**	Not reported anywhere. Missingness-as-signal exists in clinical ML, but its interaction with privacy noise does not.
8	**A genuinely paired multimodal clinical FL benchmark under patient-level DP.**	Multimodal FL papers pair modalities by construction rule or use unpaired corpora. Ours are paired by the recording apparatus — same patient, same second.

*Honest boundary: items 2–8 are **measurements and practices**, not new mathematics. The mechanisms themselves (user-level DP, parameter-efficient DP fine-tuning, DP quantile release) are prior art and are cited as such. Claiming to have invented them would be an immediate desk reject. Item 1 is the closest thing to a genuinely new scientific result.*

7 — WHAT WAS BUILT FOR THE FRAMEWORK

Component	File	What it does
Patient-level DP engine	`privacy/patient_dp.py`	The core contribution. Groups gradients by patient before clipping.
DP-SGD engine	`privacy/dp_engine.py`	Opacus wrapper with extended RDP orders so tight budgets are reachable.
Adaptive DP engine	`privacy/sadp_engine.py`	Per-group budget split. Retained as a negative result.
HFL training engine	`training/hfl_trainer.py`	The three-tier training loop: device → edge → cloud.
Encoder pretraining	`training/pretrain_encoders.py`	Non-private pretraining, with a deterministic mode and a public-data mode.
Baseline suite B0–B5	`training/baselines.py`	Centralised, local-only, FedAvg, FedProx, DP-FedAvg, MOON.
Multimodal model	`architecture/hfl_mm_model.py`	Two encoders plus a fusion head.
Signal encoder	`architecture/fedmamba_hc.py`	Multi-scale CNN + BiGRU + attention, DP-compatible.
Gradient compressor	`utils/compression.py`	Top-*k* + 8-bit + error feedback. 20× compression.
Conformal prediction	`utils/fedconform_hc.py`	Coverage guarantee with DP quantile release.
LiRA attack suite	`scripts/lira_attack.py`	Four attack variants, shadow models, per-patient Gaussians. The audit instrument.
***k*-scaling driver**	`scripts/k_scaling.py`	Runs the headline experiment across *k* and both arms.
Attribution	`scripts/attribution.py`	Integrated gradients, with a separate seed-stability check.
ONNX export + benchmark	`deploy/`	Edge deployment artifact and 1000-run latency measurement.
NS-3 integration	`scripts/ns3_phase5_integration.py`	Packet-level network measurement of every round.
CloudSim simulation	`hfl/HFLCloudSimulation.java`	Edge-cloud energy and scheduling model.
Objective checker	`scripts/check_objectives.py`	Resolves every objective from measured artifacts. No hardcoded numbers.
Statistics	`scripts/run_statistics.py`	Paired tests and confidence intervals for all claims.

Roughly 20 modules plus two simulators. The whole pipeline runs from one driver, and every reported number is resolved from a measured artifact rather than typed in.

8 — CONTRIBUTION OF MULTI-DATA AND MULTI-MODEL

8.1 Multi-data — two datasets

Using two very different corpora is what let us separate **system claims** from **task claims**.

- **VitalDB** (clinical, 6,257 patients, ~8 windows each) carries every privacy claim. It is the only one that can, because patient-level work needs many records per person.
- **WEAR** (wearable, 22 participants) carries the systems claims and proves they are not an artefact of one topology or payload size.

The two agree on everything structural: uplink reduction, energy, latency, reliability and privacy budget all hold on both. That is the evidence that the architecture generalises.

They disagree on the task claims, and that disagreement is itself a finding: fusion helps on VitalDB (+0.0370 AUROC) and is pointless on WEAR (+0.0004, ceiling). We report this rather than hide it.

8.2 Multi-model — two modalities

Each dataset pairs two genuinely different signal types recorded from the same person at the same moment:

Dataset	Modality A	Modality B	A only	B only	Fused	Gain
VitalDB	Intraoperative signals (time series)	Preoperative clinical + labs (static vector)	0.8283	0.7680	**0.8653**	**+0.0370**
WEAR	Accelerometer (time series)	Egocentric video features	0.9976	0.9957	**0.9980**	**+0.0004**

AUROC. VitalDB fusion gain is 5.9× the seed standard deviation over 5 seeds. WEAR is at ceiling and cannot evidence the claim.

Why the multimodal design matters to the privacy question. This is the part that is easy to miss. Two modalities mean the model has two sources of evidence about the same patient. That makes the patient — not the window — the natural unit of both *learning* and *disclosure*. A single patient's identity is carried by the combination of their signals and their clinical record. Protecting one window of one modality protects almost nothing about that person.

So multimodality is not decoration here. It is the reason the unit-of-privacy question has teeth in the first place.

> **Honest limitation**
> The fusion claim rests on **one dataset**. WEAR structurally cannot corroborate it. This is the second-strongest reason to add a third clinical dataset — the first being the *k*-law.

9 — WHY WE WANT PHYSIONET DATA (EICU-CRD)

9.1 The problem with what we have

Every privacy claim in this research rests on **one dataset from one hospital**. VitalDB comes from a single institution, and our nine "clients" are a partition we created ourselves. WEAR cannot help — it has 22 people, one client each, and no patient contributes enough records for the *k*-law to even be defined.

A reviewer will ask one question, and it is the right question:

> *"Is this a property of patient-level private federated learning,*
> *or a property of your dataset?"*

Right now we cannot answer it.

9.2 Why eICU-CRD specifically

Property of eICU-CRD	Why we need it
208 real hospitals	Our clients would be actual institutions, not a partition we invented. This is what federated learning is supposed to model, and we currently only simulate it.
Independent patient population	Different demographics, different care patterns, different equipment. If the *k*-law survives that, it is structural.
Same patient → many records structure	The *k*-law needs it. VitalDB gives about 8 records per patient; eICU gives far more, which stretches the curve into a range we cannot currently reach.
Same modality structure — time-series vitals plus static clinical variables	No new encoder needed. The existing pipeline applies almost unchanged, so this is weeks of work, not months.
Credentialed access with ethics training	Reviewers trust PhysioNet. Access requires CITI training and a verified reference, which is itself evidence the work was conducted properly.
Widely used benchmark	Results on eICU can be compared against other published work. Results on a private partition cannot.

9.3 What we will do with it — and the one rule

We will **not** simply run the same experiment and add another table. That is worth little.

Instead: fit the *k*-curve on VitalDB, write down a numeric prediction of what the gap should be at eICU's *k*, **freeze that prediction before touching the data**, then test it.

> **Why the prediction must be frozen first**
> The value of this experiment is that it **could fail**. If we fit the curve, look at eICU, and then write the prediction, no reviewer can tell that apart from retrofitting — and neither can we. This is why the current wait for access is useful time rather than lost time: the pre-registration has to be written now.

9.4 Both outcomes are publishable

If the prediction holds — the cost of patient-level privacy is structural, and we can give practitioners a rule: *given your k, this is what ε = 1 will cost you*. Nobody has published that.

If the prediction misses — we have found the boundary condition, with a dated forecast proving the story was not adjusted after the fact. That is also publishable, and visibly honest.

A paper reporting that two independent clinical datasets were tested, one confirming and one exposing a limit, is stronger than one where everything conveniently worked.

9.5 Current status

PhysioNet credentialing is in progress. The guide has been emailed to act as reference. Nothing else in the project is blocked by this — the manuscript restructuring, the pre-registration, and the eICU pipeline itself can all be written before access arrives.